

Expt 12

FIR Filters using DSP Processor

12.1 Design and Implementation of FIR High Pass filter using Rectangular Window

Objectives:

1. To design an FIR filter using the Rectangular Window method with specified filter requirements.
2. To implement the designed FIR filter on the TMS320C5416 DSP Starter Kit.
3. To apply an input signal to the FIR filter and obtain the filtered output using the DSP processor.
4. To analyze and verify the performance of the FIR filter by observing the input and output responses on the TMS320C5416 Starter Kit.

Hardware and Software Required:

DSP – TMS320VC5416-5416 Board

Cpu

ME Universal Debugger DSP – 5416

Procedure:

1. Connect the serial port cable between the PC and the VSK-5416 kit.
2. Switch ON the VSK-5416 kit and press the RESET button.
3. Open Vi Universal Debugger.
4. Select Serial → Port Settings → Auto Detect and verify that the message "Vi Universal Debugger found in COMx" appears.
5. Create a new project (File → New Project) and assign a project name.
6. Create a new source file (File → New File) and write the program.
7. Save the source file.
8. Add the source file to the project.
9. Add the command file (Micro5416.cmd) to the project.
10. Save the project.
11. Build/Rebuild the project and ensure there are no errors.
12. Download the executable file to the VSK-5416 kit.
13. Run the program.

14.Observe and verify the output.

HIGH PASS FIR FILTER:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Filter Specifications
```

```
Fs = 43000;      % Sampling frequency = 43 kHz
```

```
Fc = 2000;       % Cut-off frequency = 2 kHz
```

```
N = 52;         % Filter order
```

```
%% FIR High-Pass Filter using Rectangular Window
```

```
wc = Fc/(Fs/2);
```

```
b = fir1(N, wc, 'high', rectwin(N+1));
```

```
%% Display Coefficients
```

```
disp('--- FIR High-Pass Rectangular Window Coefficients ---');
```

```
disp(b');
```

```
%% Frequency Response
```

```
figure;
```

```
freqz(b, 1, 1024, Fs);
```

```
title('52nd Order FIR High-Pass Filter - Rectangular Window');
```

```
grid on;
```

```
%% Test Signal
```

```
t = 0:1/Fs:0.01;
```

```
% 1 kHz + 5 kHz test signal
```

```
x = sin(2*pi*1000*t) + 0.8*sin(2*pi*5000*t);
```

```
%% Apply FIR Filter
```

```
y = filter(b, 1, x);
```

```
%% Input and Output
```

```
figure;
```

```
subplot(2,1,1);
```

```
plot(t, x);
```

```
title('Input Signal');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
subplot(2,1,2);
```

```
plot(t, y);
```

```
title('Filtered Output');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
%% Input and Output Spectrum
```

```
figure;
```

```
subplot(2,1,1);
```

```
pwelch(x, [], [], [], Fs);
```

```
title('Input Signal Spectrum');
```

```
subplot(2,1,2);
```

```
pwelch(y, [], [], [], Fs);
```

```
title('Output Signal Spectrum');
```

Output:



Result:

1. The FIR filter was successfully designed using the Rectangular Window method.
2. The designed filter was implemented and executed on the TMS320C5416 DSP Starter Kit.
3. The input signal was processed through the FIR filter, and the filtered output was obtained.
4. The output response verified the desired filtering operation, demonstrating the successful implementation of the FIR filter on the DSP processor.